Master Programme in Generative AI and Agentic AI — Complete Curriculum

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Complete Content from DV Analytics

Course Overview

This is a focused master programme for professionals who want to specialise in **Generative AI, Agentic AI, and LLM deployment**. It begins with Python foundations, covers essential Deep Learning (with NLP focus), and progresses to building and deploying production-grade GenAI and Agentic AI systems on AWS.

Prerequisites

Before joining this programme, students should be comfortable with:

- Basic understanding of programming concepts
- Familiarity with data concepts *(optional but helpful)*

> **No prior ML or AI experience required** — this programme builds from scratch.

Programme Structure at a Glance

Module	Title
Module 1	Python for AI Development
Module 2	Basics of Deep Learning — NLP Focus
Module 3	Generative AI
Module 4	Agentic AI
Module 5	LLMOps Deployment on AWS

Module	Title
Module 6	GenAI and Agentic AI Projects — End to End

Module 1 — Python for AI Development

Section 1 — Python Fundamentals

- Introduction and Setup
- Variables and Data Types
- Operators and Control Flow — `if`, `for`, `while`
- Functions — arguments, return values, lambda
- Error Handling — `try`, `except`, `finally`

Section 2 — Data Structures

- Lists, Tuples, Dictionaries, Sets
- String operations and formatting
- List comprehensions and generators

Section 3 — File Handling and APIs

- File Operations — read, write, append
- Working with JSON and CSV
- Calling REST APIs with `requests` library
- Environment variables and secrets management

Section 4 — NumPy and Pandas for AI

- **NumPy** — arrays, operations, random
- **Pandas** — DataFrame, read CSV, clean, transform, aggregate
- **Data Visualization** — Matplotlib and Seaborn basics

Module 2 — Basics of Deep Learning — NLP Focus

Section 1 — Neural Network Foundations

- What is a neural network — perceptron analogy
- Activation Functions — ReLU, Sigmoid, Softmax
- Forward Propagation and Backpropagation — intuition
- Loss Functions — Cross-Entropy, MSE
- Optimizers — SGD, Adam
- Regularization — Dropout, L2
- Framework — Keras and PyTorch basics

Section 2 — Natural Language Processing

- Text Preprocessing — tokenization, stemming, lemmatization, stopwords
- Text Representation — Bag of Words, TF-IDF
- Word Embeddings — Word2Vec, GloVe
- Recurrent Neural Networks (RNN) — sequence modelling
- Long Short-Term Memory (LSTM) — solving vanishing gradient
- Gated Recurrent Unit (GRU)
- Bidirectional RNNs
- Encoder-Decoder Architecture
- Attention Mechanism — what it solves

Section 3 — Transformer Architecture

- Why Transformers replaced RNNs
- Self-Attention Mechanism — how it calculates relevance
- Multi-Head Attention
- Positional Encodings
- Encoder-Decoder Structure
- Tokenization — BPE, WordPiece, SentencePiece
- **BERT** — Bidirectional Encoder — used for understanding text
- **GPT** — Decoder-only — used for generating text

- **T5** — Text-to-Text — unified framework
- Hugging Face Transformers — practical usage
- Transfer Learning and Fine-Tuning concepts

Module 3 — Generative AI

Section 1 — Generative AI Fundamentals

- What is Generative AI — definition and core concepts
- Classical ML vs Deep Learning vs Generative AI
- Generative Model Taxonomy — GANs, VAEs, Diffusion Models, Transformer-based
- Language Model Capabilities and Limitations — reasoning, emergent behaviours, failure modes
- Open vs Closed-Source LLM Licenses
- Hugging Face Hub and Model Repositories — selecting and downloading models

Section 2 — Large Language Models (LLMs)

- LLM Architecture Components — layers, hidden dimensions, pre-training
- Context Window and Memory Management
- Sampling and Generation — temperature, top-p, top-k, beam search
- Alignment Techniques — RLHF, DPO, Constitutional AI
- Advanced Architectures — Mixture of Experts (MoE)
- Model Optimization — Quantization (INT8, INT4), Pruning, Knowledge Distillation

Section 3 — Prompt Engineering

- Prompt Fundamentals — structure, system messages, user messages
- Zero-shot, Few-shot, One-shot prompting
- Role-playing and personas
- Chain-of-Thought (CoT) — step-by-step reasoning
- Tree-of-Thought
- Self-consistency
- Structured Outputs — JSON mode and schema enforcement

- Prompt Injection Defense
- Prompt Security and Optimization
- LangSmith for prompt debugging and versioning

Section 4 — Retrieval-Augmented Generation (RAG)

- What is RAG — why models need external knowledge
- RAG Architecture — retriever and reader
- When to use RAG vs fine-tuning
- Embeddings and Similarity — embedding models, cosine similarity
- Vector Databases — FAISS, Pinecone, Weaviate, Chroma, Milvus, Qdrant
- Document Processing — text extraction, chunking strategies, metadata
- Semantic Search and Hybrid Search
- Re-ranking Mechanisms
- Advanced RAG Patterns — query decomposition, HyDE, multi-hop reasoning
- Graph RAG — knowledge graphs and entity linking
- Context Optimization — context window management, lost-in-the-middle
- RAG Evaluation — RAGAS, ARES frameworks

Section 5 — Multimodal AI

- Multimodal Fundamentals — cross-modal understanding
- Vision-Language Models — CLIP, BLIP, LLaVA
- Image Generation and Editing — Stable Diffusion, DALL-E
- Audio Processing — Whisper (speech recognition), text-to-speech
- Video Understanding
- Multimodal RAG — image-text retrieval
- Document Understanding — PDF parsing, table extraction, layout analysis
- Multimodal Prompting — combining text and images

Section 6 — Fine-Tuning and Model Optimization

- Fine-Tuning Fundamentals — transfer learning, when to fine-tune
- Dataset Preparation — collection, cleaning, format standardization

- Parameter-Efficient Fine-Tuning — **LoRA** and **QLoRA**
- Adapter Layers and Prefix Tuning
- Instruction Tuning and Dataset Creation
- Supervised Fine-Tuning (SFT)
- Alignment Fine-Tuning — RLHF, PPO, DPO
- Catastrophic Forgetting and Mitigation
- Knowledge Distillation — teacher-student framework
- Model Merging — weight averaging, task arithmetic

Section 7 — Safety and Guardrails

- AI Safety Principles
- Content Moderation — toxicity detection, hate speech, NSFW filtering
- Input Validation and Sanitization
- Output Validation — harmful content, PII detection
- Prompt Injection Defense
- Jailbreak Prevention — system message hardening
- Hallucination Mitigation — uncertainty quantification, Chain-of-Verification
- Constitutional Constraints
- Safety Classifiers
- Human Oversight — human-in-the-loop for high-risk decisions
- Red-Teaming — OWASP Top 10 for LLMs

Section 8 — Evaluation and Testing

- Defining Success Criteria and Metric Selection
- Offline Evaluation — static dataset, ground truth comparison
- Online Evaluation — A/B testing, canary deployment
- LLM-as-Judge — using LLMs to evaluate other LLMs
- RAG-Specific Evaluation — retrieval and generation metrics
- Human Evaluation — crowdsourcing, expert evaluation

Module 4 — Agentic AI

Section 1 — Agentic AI Foundations

- AI agents vs models — what makes an agent
- Autonomous vs semi-autonomous agents
- Perception-Action Loop — observe, decide, act, feedback
- Planning and Reasoning — ReAct paradigm, Reflexion
- Tool Use and Function Calling
- Multi-step Task Execution
- **Agent Memory Systems:**
  - Short-term memory — context window, conversation history
  - Long-term memory — vector stores, database storage
  - Episodic memory — event sequence storage
  - Semantic memory — fact storage, knowledge graphs
- Agent State Management — stateful vs stateless

Section 2 — Agent Architectures

- Single-Agent Systems — ReAct agents and tool-augmented agents
- ReAct Agent Implementation — Thought-Action-Observation loops
- Multi-Agent Systems — coordination and communication
- Hierarchical Agents — high-level planning and low-level execution
- Manager-Worker Patterns — task allocation strategies
- Collaborative vs Competitive Agents
- Advanced Patterns — meta-learning, swarm intelligence

Section 3 — Agentic Frameworks

- **LangChain** — LCEL, chains, memory management, tool development
- **LangGraph** — stateful workflows, graph-based flows, checkpointing
- **AutoGen** — conversable agents, group chats, multi-agent orchestration
- **CrewAI** — role-based agent design, task delegation, collaboration
- Framework Comparison — LangChain vs LangGraph vs AutoGen vs CrewAI

- Human-in-the-Loop Integration — approval workflows, human feedback

Section 4 — Multi-Agent Systems

- Multi-Agent Architecture — centralized vs decentralized control
- Communication Protocols — message passing, pub-sub patterns
- Agent Routing and Orchestration — dynamic routing, semantic routing
- Role Design and Delegation — RACI matrix, task decomposition
- Coordination Mechanisms — consensus building, conflict resolution

Section 5 — Agent Memory and Persistence

- Memory Architecture Design — multi-tier memory hierarchy
- Short-Term and Long-Term Memory Implementation
- Memory Operations — encoding, consolidation, retrieval, updating
- Memory Summarization — compressing long conversations
- Cross-Session Persistence — session management, privacy
- State Management — SQLite, Redis, PostgreSQL, checkpointing

Section 6 — Real-Time and Event-Driven Agents

- Event-Driven Architecture — event sourcing, pub-sub patterns
- Webhooks — fundamentals, security, retry logic
- FastAPI for Agent APIs — REST design, Pydantic models
- WebSocket Connections — real-time bidirectional communication
- Background Tasks and Async Processing
- Streaming Responses — Server-Sent Events, streaming LLM outputs
- External Integrations — Gmail, Google Calendar, Slack, CRM tools

Section 7 — Observability and Monitoring for Agents

- Observability Fundamentals — metrics, logs, traces
- **LangSmith** Platform — dataset management, prompt versioning, A/B testing
- **Helicone** Analytics — request logging, cost tracking, latency analysis
- **Grafana** Dashboards — metrics visualization and alerting

- Tracing and Debugging — distributed tracing, performance bottlenecks

Section 8 — Model Context Protocol (MCP)

- MCP Fundamentals — protocol, core architecture, MCP vs traditional APIs
- MCP Server Development — creating servers, tool registration, schema definition
- MCP Client Integration — connecting clients, tool discovery, error handling
- LLM Connectors — connecting LLMs to external systems
- MCP Security — authentication, permission and access control
- MCP Deployment — containerization, orchestration, monitoring

Section 9 — Specialized Agent Patterns

- SQL Agents — natural language to SQL, query optimization, security
- Code Analysis and Review Agents — static analysis, security scanning
- Test Generation Agents — unit tests, integration tests, edge cases
- Documentation Agents — code docs, API docs, README creation
- Research and Synthesis Agents — multi-source gathering, literature review
- Data Curation Agents — quality assessment, cleaning, schema inference
- Recommendation Agents — collaborative, content-based, hybrid
- Planning and Scheduling Agents — calendar management, resource allocation

Section 10 — Domain-Specific Applications

- **Finance and Banking** — fraud detection, risk assessment, portfolio management
- **Healthcare and Clinical** — clinical note summarization, diagnostic support, HIPAA
- **Legal Applications** — contract analysis, legal research, e-discovery
- **Retail and E-commerce** — product recommendations, dynamic pricing
- **Technology** — code generation, automated code review, test generation
- **Education and Tutoring** — personalised learning paths, homework help
- **Business Intelligence** — natural language to SQL, automated reports
- **Content Creation** — blog posts, social media, SEO, brand voice
- **Research** — literature review automation, grant writing support

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Module 5 — LLMOps Deployment on AWS

Section 1 — LLMOps Introduction

- What is LLMOps — LLM lifecycle in production
- Differences between MLOps and LLMOps
- Challenges unique to LLM deployment — latency, cost, hallucinations
- LLMOps stack — development, deployment, monitoring, evaluation

Section 2 — Containerizing LLM Applications

- Docker for LLM apps — packaging model + API + dependencies
- Writing Dockerfiles for FastAPI LLM endpoints
- Multi-stage builds for smaller images
- Docker Compose for local development
- Amazon ECR — private Docker registry on AWS

Section 3 — Deploying LLMs on AWS

- **Amazon Bedrock** — managed LLM service — Claude, Titan, Llama
- **SageMaker JumpStart** — deploy foundation models with one click
- **SageMaker Endpoints** — custom model serving with auto-scaling
- **Amazon EKS** — Kubernetes deployment for LLM APIs
- **AWS Lambda** — serverless LLM inference for event-driven workloads
- FastAPI on EKS — production REST API for LLM

Section 4 — CI/CD for GenAI Applications

- GitHub Actions for LLM app deployment
- Automated testing for LLM outputs — evaluation in CI/CD
- Canary deployment for new LLM versions
- Rollback strategies when new LLM degrades
- GitHub Secrets for API keys — OpenAI, Anthropic, AWS

Section 5 — Monitoring LLMs in Production

- **LangSmith** — tracing every LLM call, prompt versioning, debugging
- **Helicone** Analytics — token usage, cost per request, latency distribution
- **Prometheus** metrics for LLM API
- **Grafana** dashboards — visualize token cost, latency, error rate
- **Amazon CloudWatch** — infrastructure monitoring and alarms
- SNS alerts — email when cost exceeds budget or errors spike

Section 6 — Cost Optimization for LLMs

- Token usage tracking and budgeting
- Model selection — GPT-4 vs Claude vs Llama cost comparison
- Caching strategies — exact match and semantic caching
- Batch inference vs real-time — when to use each
- Prompt compression — reduce tokens without losing meaning
- Spot instances for training and fine-tuning

Section 7 — LLM Evaluation in Production

- Defining evaluation metrics for your use case
- Offline Evaluation — static dataset, ground truth comparison
- Online Evaluation — A/B testing prompt versions
- LLM-as-Judge — Claude evaluates GPT outputs
- RAG Evaluation — RAGAS framework — faithfulness, relevancy, context precision
- Agent Evaluation — task completion rate, tool usage appropriateness
- Human Evaluation — when automated evaluation is not enough

Section 8 — Security for LLM Applications

- API key management — AWS Secrets Manager
- Input sanitization — prevent prompt injection in production
- Output filtering — PII detection, harmful content
- Rate limiting and abuse prevention
- VPC deployment for sensitive LLM applications
- Audit logging — who called what, when, with what prompt

Module 6 — GenAI and Agentic AI Projects — End to End with Deployment

Project 1 — Document Intelligence System (RAG)

What it does:
Users upload PDF documents — financial reports, legal contracts, medical records. The system extracts text, chunks it, embeds it, and stores in a vector database. Users ask questions in natural language and get accurate answers with source citations.

Industry Use Cases: Law firms | Finance | Healthcare | HR

Tech Stack:

- LangChain RAG pipeline
- Chroma vector database
- OpenAI or Claude for LLM
- FastAPI REST endpoint
- Streamlit UI for document upload and Q&A
- Docker + ECR + EKS deployment
- LangSmith monitoring and tracing

Project 2 — Multi-Agent Research and Report Generator

What it does:
User gives a research topic. Multiple specialised agents work together — one searches the web, one analyses sources, one writes the report, one reviews and edits. Final output is a structured research report in minutes.

Industry Use Cases: Market research | Consulting | Investment banking | EdTech

Tech Stack:

- CrewAI or LangGraph multi-agent system
- Researcher agent — web search tool
- Writer agent — structured report generation
- Reviewer agent — quality check and citation verification

- Streamlit interface for topic input and report download
- FastAPI backend
- Docker + ECR + EKS deployment
- LangSmith observability

Project 3 — Intelligent Customer Support Agent

What it does:
A customer support agent that handles queries, accesses customer history, resolves common issues automatically, and escalates complex cases to human agents — with full memory across sessions.

Industry Use Cases: E-commerce | Banking | Telecom | Healthcare

Tech Stack:

- LangChain or LangGraph agent
- Tools — CRM lookup, order status, knowledge base search
- Memory — SQLite for conversation history across sessions
- FastAPI REST endpoint
- Streamlit or web chat interface
- Docker + ECR + EKS deployment
- Helicone cost tracking
- CloudWatch monitoring

Project 4 — SQL Analytics Agent

What it does:
Business users ask questions in plain English. The agent translates to SQL, queries the database, generates insights, creates visualisations, and presents the answer in a readable format — no SQL knowledge required.

Industry Use Cases: Sales teams | Marketing | Finance | Any company with a database

Tech Stack:

- LangChain SQL Agent
- PostgreSQL or SQLite database
- Natural language to SQL with validation
- Matplotlib/Plotly for visualisations
- FastAPI endpoint
- Guardrails — prevent DELETE, DROP queries
- Streamlit business dashboard
- Docker + ECR + EKS deployment

Project 5 — GenAI Content Pipeline with Agents

What it does:
An agentic pipeline that generates, reviews, and publishes content automatically. Given a topic — the system researches it, writes a blog post, creates social media variations, checks for brand tone, and schedules for publishing.

Industry Use Cases: Marketing agencies | EdTech | News organisations | Product companies

Tech Stack:

- LangGraph workflow agent
- Research agent — web search
- Writing agent — long-form content
- Social agent — short-form variations
- Review agent — brand voice check, hallucination detection
- Integration with CMS or email via API
- FastAPI + Docker + ECR + EKS
- LangSmith full pipeline tracing

Tools and Technologies Covered

Category	Tools
Programming	Python
Data	Pandas, NumPy
Deep Learning	PyTorch, TensorFlow/Keras
NLP	Hugging Face Transformers, NLTK, spaCy
Generative AI	OpenAI GPT, Anthropic Claude, Google Gemini
LLM Frameworks	LangChain, LangGraph, AutoGen, CrewAI
RAG	FAISS, Pinecone, Chroma, Weaviate, Qdrant
Prompt Engineering	LangSmith, structured outputs
Fine-Tuning	LoRA, QLoRA, PEFT, Hugging Face TRL
Vector Databases	Chroma, Pinecone, Weaviate, FAISS, Milvus
Cloud	AWS — Bedrock, SageMaker, ECR, EKS, Lambda, CloudWatch
Monitoring	LangSmith, Helicone, Prometheus, Grafana, CloudWatch
CI/CD	GitHub Actions
Containerisation	Docker
API Framework	FastAPI, Streamlit
MCP	Model Context Protocol servers and clients
Security	AWS Secrets Manager, input/output guardrails

What You Will Build

By the end of this programme, you will have built and deployed:

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✅ Document Q&A system           – RAG with vector database
✅ Multi-agent research assistant – CrewAI/LangGraph
✅ Customer support agent         – with memory and CRM integration
✅ SQL analytics agent            – natural language to database
✅ Content generation pipeline    – full agentic workflow
```

All deployed on AWS with:

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✅ Docker containers on EKS
✅ CI/CD pipeline via GitHub Actions
✅ Full observability — LangSmith + CloudWatch
✅ Cost monitoring  — Helicone
✅ Security         — API keys, rate limiting, guardrails
```

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